

Modeling & generation

Describe the business once in AD Vestro. Everything else is built on that model.

AD Vestro

Visual entity designer — generates all boilerplate files as a ZIP.

AI-Assisted Entity Design

A plain-English description generates a reviewable draft entity directly in AD Vestro. It never applies or generates code on its own.

[PlatformField] Attribute

Decorate entity properties to drive automatic UI generation.

LookupFk / SoftLink

FK and autocomplete inputs — any entity can be a lookup target.

LookupDisplayField

Use any property as the display field for a LookupFk, not just Name.

Many-to-Many Grid

AutoManyToManyGrid for junction-table relationships. AD Vestro generates the junction entity automatically.

Generated Sequences

Auto-incrementing prefixed values (e.g. "REF: 00001") configured entirely in AD Vestro, with the hook auto-generated in the ZIP.

IEntityRegistry

Generic entity registry, auto-populated by AddEntity. Drives Reports and future features.

AddEntity<T> extension

One-line repository registration — auto-registers all the interfaces an entity needs.

Automatic UI

Lists, forms, grids and filters generated from the model, with the behaviour applications repeatedly need.

AutoListPage

List page with filters, export, import, bulk delete, saved views and notifications.

AutoDetailPage

Detail/edit page with FluentValidation, workflow and child tabs.

AutoForm

Auto-generated form with FluentValidation — inline errors and a summary.

AutoGrid

Server-side grid with LookupFk resolution, checkboxes and an empty state.

AutoFilter

Filter bar driven by [PlatformField] metadata.

AutoChildGrid

Inline child grid for master/detail — auto-hides the parent FK field.

Saved Views

Save and restore grid state — filters, sort, search and page size.

Bulk Delete

Select multiple rows and delete in one action.

Dashboard Widgets

StatCard, BarChartWidget, DonutChartWidget and RecentItemsWidget.

Layout Theming

MudTheme and LayoutSettings via appsettings.json.

Dark Mode

Per-user dark/light mode stored in the database via UserPreferenceState.

Multi-Language Support

resx for static text and a DB-backed TextTranslation table for dynamic AD Vestro-defined labels, both behind one ILocalizationService, with a per-user Language preference.

Data, documents & reporting

Get data in, get it out, and attach the files that belong to a record.

Generic Import

CSV and Excel import with column mapping, preview and validation.

ExportButtons

PDF and Excel export driven by ShowInExport metadata.

Report Designer

Create, run and export grouped/aggregated reports with PDF and Excel output.

Record PDF Generation

IRecordPdfBuilder — one-record work orders, invoices and similar client-facing PDF documents.

Image Management

Generic entity image upload and gallery via IHasImages — auto-tab on AutoDetailPage, secure storage.

Document Management

Versioned file uploads and downloads per entity via IHasDocuments — configurable categories, secure storage.

Generic API Layer

JWT-authenticated REST API.

Business logic & data integrity

Extension points and safeguards that keep the application correct as it grows.

Workflow Engine

State machine — no-code via the Workflow Designer, or in code via `IWorkflowDefinition<T>` — with DB-stored permissions, history and comments.

Hooks (Entity + Workflow)

Custom logic attachment points: CRUD lifecycle (BeforeAdd, AfterAdd, BeforeUpdate, AfterUpdate, BeforeDelete, AfterDelete) plus workflow transitions.

Concurrency Management

Optimistic concurrency via `IConcurrencyAware/RowVersion`, with a Reload/Overwrite/Cancel conflict dialog.

Entity Lifecycle Support (Soft Delete)

Soft delete, restore, hard delete and purge cleanup via `ISoftDeletable`.

GenericFilteredRepository

Base repository with search, sort, filter and audit.

GenericLookupRepository

Base for lookup entities — extends `GenericFilteredRepository` and `ILookupRepository`.

Notifications & communication

Reach users in the app, by email, or on their device.

Notification System

SignalR real-time with bell icon, badge, toast, history and optional per-user email delivery.

Web Push Notifications

Real OS-level push, delivered even with the browser closed — VAPID-based, per-user opt-in, sharing the same recipient model as the bell and email channels.

Scheduled Notifications

Background service for user reminders and application jobs.

User Reminders

Self-service scheduled reminders — OneTime, Daily, Weekly, Monthly or Cron — delivered via the standard notification system.

Email Log

Admin-viewable history of every email delivery attempt, independent of console output.

User Presence & Directory

IUserPresenceService and IUserDirectoryService — who is currently connected, and the full user list, for any app-level user picker.

Cleanup Jobs

Auto-delete old read notifications and audit logs on a schedule.

Security & administration

Identity, access and accountability from the first run.

User Management

Full CRUD with ASP.NET Core Identity.

Role Management

Roles with an IsAdmin flag and per-module permissions.

Permission System

Per-role, per-module CRUD permissions.

Module System

Dynamic navigation menu with nesting via ParentGroup.

Audit Logging

Automatic EF Core interceptor logging all database changes.

Secrets Vault

Encrypted, per-machine storage for API keys and signing secrets — Admin → Secrets, never appsettings.json or environment variables.

User Preferences

Per-user settings in the database. Extensible.